

Cultural Matching and the Persistence of Social Ties

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Abstract

How does cultural matching shape the evolutionary dynamics of personal networks during major life transitions? While network theories frequently emphasize the structural and demographic drivers of tie dynamics, I advance an integrative framework in which cultural matching actively protects personal ties from decay alongside structural opportunity. Using longitudinal ego-network data from the *NetSense* study, which tracked college students across eight survey waves, I find that cultural matching—both in broad categorical domains and in specific leisure activities—significantly reduces the hazard of tie decay, net of dyadic structural embeddedness (triadic closure), residential proximity, and demographic homophily. Conversely, I find that *cultural network opacity*, or a lack of mutual cultural knowledge, acts as an independent relational liability accelerating tie decay. Crucially, the protective effect from tie decay via cultural matching is heavily concentrated among subjectively close ties: cultural matching forms an essential interactional core for maintaining close relationships amidst high background churn, whereas weak ties face such a high baseline rate of decay that cultural matching alone does little to sustain them. These findings show that, within high-turnover network ecologies, individuals draw upon cultural matching as a portable resource to selectively curate and protect their most meaningful social bonds from decay.

1 Introduction

A central issue in the study of culture and networks concerns the co-evolution of social ties and cultural tastes (Lizardo, 2006; Edelman and Vaisey, 2014). Contemporary research in network science and computational social science has shown that relational structures and cultural practices evolve via dynamic, reciprocal feedback (Mark, 1998b; McPherson, 2004; Lewis and Kaufman, 2018). Social ties provide the structural channels and interactional opportunities through which cultural information and practices diffuse and are subsequently adopted (Lazarsfeld and Merton, 1954; Carley, 1991; Borgatti, 2005), and cultural profiles guide partner selection and tie formation, generating choice and status homophily thus reinforcing network boundaries (DiMaggio, 1987; McPherson et al., 2001; Shalizi and Thomas, 2011).

Yet a long-standing finding in ego-network research is that personal networks are characterized by pervasive turnover, with ties constantly being formed, renegotiated, and pruned—particularly during critical life-course transitions such as matriculating into college, getting married, having children, changing jobs, relocating, or retiring from paid work (Burt, 2000; Wellman et al., 1997; Kleinbaum, 2018; Roy et al., 2022; Weiss et al., 2022; Kempnich et al., 2024; Vacchiano et al., 2024). This endemic personal network volatility raises a key question: amid high background churn, how do individuals selectively decide which ties to preserve and which to allow to decay, and what role do cultural matching (Vaisey and Lizardo, 2010), structural embeddedness (Burt, 2000), and institutional foci (Feld and Grofman, 2009) play in protecting personal ties from decay?

In this paper, I address this question by evaluating the empirical implications of an *integrative cultural matching model* of tie decay within the dynamic setting of a sample of students during the transition to college. Rather than treating cultural tastes as passive byproducts of network positions or framing culture and structure as competing explanations, I examine how cultural matching operates in tandem with structural opportunity structures to modulate the dynamics of tie durability (Burt, 2000; Lewis and Kaufman, 2018; Rivera et al., 2010). Under this framework, dyadic cultural matching serves as a vital micro-interactional anchor, increasing protection from tie decay, reducing interactional friction, generating positive emotional energy in shared conversational rituals (Collins, 2004; Lizardo, 2016), elevating the cognitive salience of alters, and ultimately rendering relationships resilient against background turnover (DiMaggio, 1987; Lizardo, 2006; Meuleman, 2021; Meuleman and Jæger, 2023). Crucially, I examine cultural matching not in a structural vacuum, but alongside key dimensions of positional and collective embeddedness—including triadic closure (shared contacts in the ego network), spatial foci (residential dorm co-presence), subjective closeness, and sociodemographic homophily.

1.1 Empirical Context

I use rich longitudinal ego-network panel data from the *NetSense* study across eight survey waves (Bahulkar et al., 2016, 2017; Hachen et al., 2022), tracking the decay and persistence of non-kin, on-campus peer ties across multiple college-year transitions. This empirical context provides a strategic window into relationship dynamics during emerging adulthood, capturing both the rapid reorganization of peer networks during the freshman transition and the

subsequent consolidation of personal communities as graduation approaches. The *NetSense* design uniquely combines repeated ego-network nominations with granular measures of both broad categorical tastes and concrete, open-ended leisure activities, as well as complete alter-alter acquaintance networks that allow us to account for the structural embeddedness of each tie.

Empirically, I examine the cultural matching mechanism along three interrelated dimensions. First, I establish the baseline protective effect of *cultural matching*—evaluating both broad categorical domain matches and concrete leisure activity matches—against tie decay, adjusting for dyadic structural embeddedness and residential proximity. Second, I introduce the concept of *cultural network opacity*—the degree to which an ego lacks mutual cultural knowledge of an alter’s preferences—as an independent predictor of tie decay. Third, I investigate whether the protective capacity of cultural matching is concentrated in strong, subjectively close relationships or serves as compensatory scaffolding for weak ties to overcome what (Burt, 2000) calls the *liability of newness*. By testing these mechanisms in a discrete-time event-history framework with ego fixed effects and cross-classified random effects models, I provide robust evidence that cultural matching operates as an active mechanism that protects ties from decay in dynamic peer environments.

The remainder of this article proceeds as follows. First, I develop an integrative theoretical framework linking cultural matching, structural position, and tie decay during life transitions. Next, I describe the NetSense panel data, measures of taste stability, and the discrete-time event history modeling strategy. Finally, I present the empirical results and discuss their theoretical implications for research on homophily, network turnover, and cultural capital.

2 Theoretical Framework

2.1 An Integrative Perspective on Culture and Network Dynamics

The relationship between cultural tastes and social networks has long been a central question at the intersection of cultural sociology and structural network analysis. Early formulations often treated culture and structure as competing explanations for social association (Erickson, 1996; Mark, 1998a, 2003; Lizardo, 2006). On one side, structural perspectives emphasized how exogenous opportunity structures, physical foci of activity (Feld, 1981), and spatial propinquity dictate interaction opportunities, with tastes diffusing secondarily across homophilous ties as fluid, network-contingent contents (Carley, 1991; Borgatti, 2005). On the other side, cultural capital perspectives conceptualized tastes as durable, embodied dispositions acquired through early socialization that individuals actively deploy to select, maintain, and discard social ties (Bourdieu, 1984, 1990; Holt, 1998; Childress and Nault, 2019).

Recent advances in dynamic network modeling and digital trace research suggest that moving beyond a zero-sum opposition between culture and structure provides a more productive theoretical orientation (Lewis and Kaufman, 2018; Shalizi and Thomas, 2011; Edelmann and Vaisey, 2014). Relational dynamics are fundamentally co-evolutionary: structural opportunity structures bring individuals into contact, while cultural matching determines whether those interactions generate mutual affinity, emotional energy, and ongoing commitment (Collins, 2004; Lizardo, 2006). As Lewis and Kaufman (2018) propose in their generalized conversion

model, cultural conversion (connectivity resulting from diverse ties) and cultural matching (forming and sustaining specific ties based on convergent tastes) operate simultaneously, bounded by the local cultural and institutional ecology.

This co-evolutionary perspective is especially relevant when analyzing network evolution during major life transitions. Longitudinal studies consistently show that personal networks experience high turnover, with between one-third and two-thirds of active ties dissolving over relatively short intervals (Morgan et al., 1997; Suitor et al., 1997; Roy et al., 2022; Weiss et al., 2022). If social ties were the sole determinants of cultural tastes, individual cultural profiles would exhibit turbulence matching the rapid churn of their local networks. Yet empirical research indicates that core cultural orientations and aesthetic dispositions remain relatively stable across emerging adulthood (Vaisey and Lizardo, 2010; Childress and Nault, 2019). If personal networks are volatile while cultural dispositions are durable, cultural matching must serve as a critical sorting mechanism through which individuals actively navigate relational churn, selectively protecting culturally compatible ties while allowing mismatched or opaque connections to decay (DiMaggio, 1987; Lizardo, 2006).

2.2 Cultural Matching, Interaction Rituals, and Tie Decay

A fundamental challenge confronting newly formed social relationships is what Burt (2000), drawing on Stinchcombe (1965), terms the “*liability of newness*.” Nascent dyadic ties are fragile, characterized by ambiguous expectations, low emotional investment, and an elevated baseline hazard of decay. As relationships mature, this decay hazard typically declines, but what enables a tie to overcome this liability of newness in the first place? While structural network theories highlight network constraint, multiplexity, and third-party monitoring (Coleman, 1988; Burt, 2000), I propose that *cultural matching* provides an indispensable micro-interactional foundation that protects ties from decay.

At the micro-level, cultural matching operates through two complementary interactional mechanisms. First, drawing on Collins’s (2004) theory of interaction ritual chains, cultural matching provides the mutual conversational focal points and bodily coordination necessary to produce successful interaction rituals. When two individuals share an interest in a cultural domain—such as music, films, sports, or recreational hobbies—they can readily initiate conversation, establish shared attention, and experience mutual entrainment. These successful interaction rituals generate positive emotional energy and feelings of solidarity, making the relationship intrinsically rewarding and motivating future interaction. Conversely, when cultural matching is absent, interactions require conscious cognitive effort, feel awkward or draining, and fail to generate emotional energy, leaving the tie vulnerable to benign neglect and decay.

Second, cultural matching increases the *cognitive salience* of an alter within an ego’s personal community (DiMaggio, 1987; Burt, 2002). In high-density environments like a college campus, individuals encounter numerous casual acquaintances. Alters with whom an ego shares specific cultural pursuits remain top of mind because they serve as immediate partners for concrete leisure activities (e.g., attending concerts, playing video games, or discussing books). This cognitive accessibility makes it more likely that the alter is regularly included in social plans, protecting the tie from the out-of-sight, out-of-mind dynamics that drive relational decay. Therefore, I anticipate that dyadic cultural matching significantly reduces

the hazard of tie decay, net of structural embeddedness and dyadic controls (*Hypothesis 1*).

2.3 Cultural Network Opacity as an Interactional Liability

If shared cultural knowledge protects ties from decay, then a *lack* of mutual cultural knowledge—*cultural network opacity*—should act as an independent liability that accelerates tie decay. I define cultural network opacity as the degree to which an ego is ignorant of an alter’s cultural preferences, indicated empirically by explicit “Don’t Know” reports across cultural domains.

It is conceptually crucial to differentiate opacity from mere cultural mismatch. A cultural mismatch implies that ego and alter are aware of each other’s preferences but hold divergent tastes (e.g., one enjoys sports while the other does not). Such mismatches still represent an informed relational connection where boundaries are known. Cultural network opacity, by contrast, signifies a relational void. When an ego cannot identify an alter’s preferences across multiple basic leisure domains, it reveals a profound lack of interactional depth and personal disclosure (Wellman et al., 1997). Furthermore, opacity creates a severe cognitive barrier to initiating joint activities: without knowing what an alter enjoys, an ego cannot easily propose shared outings or anticipate the alter’s reactions. This mutual cultural ignorance renders the relationship superficial and interactionally inert, increasing the probability that the tie will be dropped from the ego’s active network. Consequently, I hypothesize that cultural network opacity independently increases the hazard of tie decay (*Hypothesis 2*).

2.4 Tie Strength Contingencies: Strong-Tie Maintenance vs. Weak-Tie Scaffolding

A central theoretical question concerns how the protective capacity of cultural matching intersects with tie strength. Existing literature suggests two competing predictions regarding this contingency.

On the one hand, drawing on cultural sociology and friendship research (Vaisey and Lizardo, 2010; Edelman and Vaisey, 2014), cultural matching may be primarily essential for the maintenance of *strong* ties. In this perspective, intimate, emotionally significant friendships demand continuous cultural matching to sustain deep conversation, mutual understanding, and authentic rapport. Because strong ties carry high cognitive and emotional stakes, any perceived cultural misalignment or opacity becomes salient and disruptive. Consequently, strong ties should be highly sensitive to cultural matching for protection from decay (*Hypothesis 3a: Strong-Tie Matching*).

On the other hand, classic structural network theory suggests an alternative prediction: strong ties may be inherently durable due to accumulated emotional inertia, institutional commitments, and dense third-party embeddedness, making them resilient to cultural differences. Under this view, cultural matching acts as essential compensatory scaffolding specifically for *weak* or newly formed ties. Weak ties suffer from a high baseline liability of newness (Burt, 2000); without the buffer of deep intimacy, they require immediate cultural commonalities to grease the wheels of interaction and justify continued contact. Once a tie matures into a close bond, cultural commonalities may become secondary to generalized interpersonal trust and shared history. Thus, the protective power of cultural matching against tie decay should be most pronounced among weak ties (*Hypothesis 3b: Weak-Tie Scaffolding*).

2.5 Positional Context, Institutional Structure, and Capital Conversion

Cultural choices and relational maintenance do not occur in an institutional or structural vacuum (Feld, 1982; McFarland et al., 2014). Actors occupy specific positional locations within campus social structures defined by residential arrangements, organizational foci, and exogenous sociodemographic identities. Incorporating positional differences is therefore essential for avoiding an essentialized view of cultural matching and for understanding the conditions under which cultural capital translates into relational durability.

First, physical and institutional foci strongly condition interaction opportunity structures. In an undergraduate setting, residential arrangements (such as living in the same dormitory) provide recurring, low-cost opportunities for face-to-face contact. Shared spatial proximity can either amplify or substitute for cultural matching: roommates and hallmates may maintain ties through spatial convenience even in the absence of cultural matching, or shared living spaces may provide the necessary stage where cultural matching can be discovered and enacted. Similarly, structural embeddedness—the extent to which an alter is connected to other members of an ego’s personal network (triadic closure)—provides reputational monitoring, collective validation, and group-level scaffolding that protects ties from decay (Granovetter, 1973; Coleman, 1988; Burt, 2000). Controlling for dyadic common alter counts and residential proximity is therefore critical to determining whether cultural matching has independent predictive power net of structural integration.

Second, drawing on Bourdieu’s (1984; 1986) theory of capital conversion, the capacity of cultural matching to convert into social capital (durable social ties) is intrinsically bounded by actors’ structural and institutional locations. Cultural capital does not convert into relational durability at an invariant exchange rate; its efficacy depends on the volume and composition of other capitals and the specific field conditions of the campus environment. Rather than operating as an unconstrained universal solvent that effortlessly dissolves social boundaries, cultural matching operates as a micro-interactional resource that functions *within* the constraints imposed by exogenous demographic categories (such as race and gender) and institutional opportunity structures (such as residential arrangements). For instance, students from minoritized racial backgrounds navigating predominantly white campus spaces face distinct structural hurdles and baseline homophily constraints that condition relational choices. For minoritized students, cultural matching may provide an important micro-interactional resource for establishing cross-group ties, yet exogenous structural boundaries and demographic isolation may impose distinct constraints on the durability of long-term ties. Incorporating exogenous demographic controls alongside structural embeddedness ensures that cultural matching is evaluated not in isolation, but net of the powerful positional boundaries that organize campus life.

2.6 Summary of Hypotheses

In sum, I evaluate four core theoretical predictions regarding culture, structure, and tie decay:

- *Hypothesis 1 (Cultural Matching)*: Dyadic cultural matching significantly reduces the hazard of tie decay, net of structural embeddedness, residential proximity, and sociodemographic controls.

- *Hypothesis 2 (Cultural Network Opacity)*: Cultural network opacity independently increases the hazard of tie decay.
- *Hypothesis 3a (Strong-Tie Matching)*: The protective effect of cultural matching against tie decay is amplified among subjectively close (strong) ties.
- *Hypothesis 3b (Weak-Tie Scaffolding)*: The protective effect of cultural matching against tie decay is amplified among weak (not close) ties to overcome the liability of newness.

3 Data and Methods

Data for this analysis comes from the *NetSense* study (Striegel et al., 2013), a longitudinal survey of college students conducted at the University of Notre Dame between 2011 and 2015. The study collected extensive panel data on students’ social networks, demographic background, and cultural preferences. The survey gathered information on the cultural habits of ego-alter pairs using both a series of closed-form domain questions and open-ended nominations of activities. For the closed-form items, egos were asked to rate both their own and their alters’ level of interest across six broad cultural and leisure domains: *music*, *movies*, *books*, *sports*, *games*, and *outdoor activities*. For the open-ended items, respondents freely nominated up to five leisure activities they enjoyed, and then indicated whether their alter also engaged in each nominated activity. Surveys were administered at the start and end of each semester via Qualtrics during the study period.

3.1 Discrete-Time Event History Setup, Risk Set, and Tie Sequences

To investigate tie decay over time, I construct a discrete-time person-period dataset spanning adjacent survey waves (Allison, 1982; Singer and Willett, 1993)—for a recent application of this approach to the study of tie evolution see Kim et al. (2024). To isolate peer dynamics, I restrict the sample to reported non-kin, on-campus ties (that is, alters who are also Notre Dame students). The NetSense study administered 8 survey waves across the students’ college careers. Information on ego cultural preferences across the six closed-form domains was gathered across Waves 1 through 6, enabling the empirical modeling of tie decay across six adjacent wave transition intervals ($t = 1 \rightarrow 2, 2 \rightarrow 3, 3 \rightarrow 4, 4 \rightarrow 5, 5 \rightarrow 6$, and $6 \rightarrow 7$). This longitudinal design allows for observing tie decay continuously across distinct developmental intervals of the undergraduate trajectory, capturing both the initial network churn of freshman year and the consolidation of personal networks as students approach upper-level student status.

The unit of analysis is the dyad-period (i, j, t) , defined for each alter j nominated in ego i ’s active interaction roster at wave t . At each wave transition interval $t \rightarrow t + 1$, an active tie is at risk of experiencing tie decay. The dependent variable, Y_{ijt} , is a binary indicator scored as 1 if the tie is nominated again by ego i at wave $t + 1$ (indicating that the tie is protected from tie decay) and 0 if the alter is omitted from the ego’s nomination roster at wave $t + 1$ (indicating tie decay). In discrete-time event history terms, the conditional hazard of tie decay at interval t is $h_{ijt} = P(Y_{ijt} = 0 \mid \text{active at } t)$, and the complementary probability

of protection from tie decay is $1 - h_{ijt} = P(Y_{ijt} = 1 \mid \text{active at } t)$. In the analytic sample, 29.5 % of dyad-periods persist into the subsequent wave, reflecting the substantial baseline rate of tie decay characteristic of college peer networks.

A methodological question arises regarding the combinatorial trajectories of ties across multiple survey waves and the treatment of intermittent ties (ties that disappear and subsequently reappear). Across 8 survey waves, a dyad’s nomination trajectory could theoretically exhibit any of $2^8 = 256$ binary presence/absence combinations ($2^7 = 128$ across the 7-wave analytical window). In the empirical NetSense data, however, only 151 distinct sequence configurations are observed. More importantly, the vast majority of dyads—79.8 % (4,716 of 5,908 unique dyads)—exhibit strictly contiguous relational histories: they are nominated in a single wave or persist continuously across consecutive waves until permanent decay or study right-censoring. Only 20.2 % of dyads (1,192 dyads) ever exhibit an intermittent pattern in which a tie is omitted in one wave and re-nominated in a later wave. Within the complete-case analytic sample ($N = 5,349$ dyad-periods across 3,770 dyads), 84.5 % ($N = 4,522$) belong to the dyad’s *first continuous spell* (from initial nomination until first decay or right-censoring), while only 15.5 % ($N = 827$) belong to recurrent spells.

Note that under this discrete-time framework, *rekindling does not count as persistence*. If an alter is nominated at wave t but omitted at wave $t + 1$, that dyad-period is coded as experiencing tie decay ($Y_{ijt} = 0$). If that alter is nominated again at a subsequent wave (for example, at wave $t + 2$), the discrete-time repeated-events model treats this re-appearance as the inception of a new, distinct risk episode (spell 2) covering the transition from $t + 2 \rightarrow t + 3$ (Allison, 1982; Box-Steffensmeier and Jones, 2004). Rekindling at wave $t + 2$ never retroactively alters the decay event observed at interval $t \rightarrow t + 1$. Furthermore, all time-varying predictors (subjective closeness, contact frequency, cultural matching, and structural embeddedness) are dynamically re-measured at wave $t + 2$ to predict the subsequent outcome at wave $t + 3$. To confirm that the findings do not depend on the inclusion of recurrent spells, I report a sensitivity analysis in Section 4.3 that strictly enforces an *absorbing first-decay* specification, tracking ties exclusively until their first decay event and dropping all subsequent rekindled spells ($N = 4,522$).

A related consideration concerns *node persistence* (ego survey retention) and the threat of survivorship or panel attrition bias. In the NetSense panel, respondents completed an average of 5.12 survey waves (median 5 waves), with 80.3 % (151 of 188 egos) participating in 4 or more waves and 48.9 % completing 6 or more waves. In discrete-time event history setups, when an ego does not complete survey wave $t + 1$, their active ties from wave t are formally right-censored at interval $t \rightarrow t + 1$. Auxiliary attrition regressions confirm that ego panel retention is completely uncorrelated with baseline cultural matching: neither closed-form matching ($p = 0.98$) nor open-ended matching ($p = 0.23$) predicts total waves completed or study dropout. To guard against potential selection biases, Section 4.3 evaluates sensitivity models restricted to complete wave-to-wave transitions and high-retention egos.

3.2 Measures and Descriptive Statistics

Tables 1 and 2 present summary statistics for all continuous and categorical variables used in the analyses across the $N = 5,584$ dyad-period observations.

Table 1: Descriptive Statistics (Continuous Variables)

	Mean	SD	Min	Max
Closed-Form Cultural Matching	1.67	1.49	0.00	6.00
Open-Ended Activity Matching	2.52	1.34	0.00	5.00
Cultural Network Opacity	0.45	1.13	0.00	6.00
Structural Embeddedness (Common Alters)	5.74	4.18	0.00	19.00
Tie Duration (Years)	4.23	6.21	0.02	21.16

Table 2: Descriptive Statistics (Categorical Variables)

Variable	Category	N	Percent
Tie Persisted (Protection from Tie Decay)	0 (Decayed)	3935	70.5%
	1 (Persisted)	1649	29.5%
Ego Gender Identity	Men	2901	54.0%
	Women	2471	46.0%
Alter Gender Identity	Men	2839	51.0%
	Women	2728	49.0%
Subjective Closeness	Not Close	565	10.1%
	Somewhat Close	2611	46.8%
	Close	2400	43.0%
Interaction Frequency	Weekly	760	13.6%
	Daily	4824	86.4%
Race Homophily	Heterophilous	510	9.1%
	Homophilous	1508	27.0%
	Unknown	3566	63.9%

3.2.1 Outcome Variable: Protection from Tie Decay

Protection from Tie Decay: Network evolution is evaluated through tie decay between adjacent waves. As detailed above, the outcome is measured as a binary indicator taking a value of 1 if a social tie reported by an ego in a baseline wave (e.g., Wave 1) is still present in the subsequent wave (e.g., Wave 2; protected from tie decay), and 0 if the tie has decayed (that is, the alter is no longer listed in the ego’s network). To determine tie presence, egos were asked to list the people with whom they interact most frequently, using the following name generator: “Please list the people with whom you have spent the most time in the last month.” Egos could nominate up to 20 alters in each wave. This name generator uses what (Marin and Hampton, 2007) calls the “interaction” approach, which is relatively neutral with respect to tie strength (Laumann et al., 1989). Other tie characteristics related to strength, such as duration, role relation (e.g., kin versus friend), and subjective closeness, were collected via follow-up name interpreters.

3.2.2 Cultural Predictors

Closed-Form Cultural Matching: A count variable (ranging from 0 to 6; Mean = 1.67, SD = 1.49) indicating the number of closed-form cultural domains in which the ego and alter exhibit cultural matching. Egos were asked to rate their own interest, and their alters’ interest across six broad domains: music, movies, books, sports, games, and outdoor activities. The specific survey prompts were: for egos, “How interested are you in the following activities?” and for alters, “How interested is this person in the following activities?” Response categories included “Very interested,” “Somewhat interested,” and “Not at all interested.” A domain was scored as a “match” if the ego rated both themselves and their alter identically across these categories. Note that the ego’s information was collected in a *separate* survey from the ego-network survey (a demographic survey), which dealt with immediate ordering bias and attenuated any cognitive consistency effects. As evaluated in Section 4.3, positive interest alignment (shared interest ratings of “Very interested” or “Somewhat interested”) accounts for 92.0 % of all observed closed-form matches.

Open-Ended Activity Matching: A count variable (ranging from 0 to 5; Mean = 2.52, SD = 1.34) capturing the degree of cultural matching across specific leisure activity nominations. Egos were asked to freely list up to 5 leisure activities they enjoyed, using the prompt: “Please list up to 5 specific activities that you enjoy doing in your free time.” Subsequently, for each alter, the ego was asked to check boxes indicating which of these specific activities the alter also enjoyed, prompted with: “Which of your favorite activities does this person also enjoy?”

Cultural network opacity: A count variable (ranging from 0 to 6; Mean = 0.45, SD = 1.13) measuring the number of cultural domains for which the ego responded “Don’t know” regarding their alter’s preferences. For the closed-form domain questions regarding alters, respondents were provided a fourth response category: “Don’t know.” While survey researchers often instinctively treat “Don’t know” responses as missing data to be imputed or dropped, I explicitly operationalize these responses as substantive data. In the context of ego-network surveys, a “Don’t know” response is not random missingness; rather, it is a direct behavioral measure of relational ignorance (opacity). By retaining and counting these responses, I treat opacity as a distinct structural feature of the dyadic tie.

3.2.3 Longitudinal Stability and Consistency of Cultural Tastes

A fundamental premise in theories of cultural capital and relational selection is that cultural tastes represent relatively durable personal orientations rather than ephemeral, fast-shifting pursuits (Lizardo, 2006). If students’ cultural tastes fluctuated erratically across semesters, or if observed dyadic similarity arose primarily because connected friends rapidly adopt and abandon new lifestyle practices in lockstep, cultural matching could not serve as an enduring micro-interactional anchor protecting ties from decay (Vaisey and Lizardo, 2010).

To evaluate the empirical durability of cultural tastes in this undergraduate population, Table 3 reports longitudinal consistency metrics across the panel. First, I estimate two-level random-intercept models ($N = 848$ observations across 170 egos) to decompose between-person and within-person variance across Waves 1–3 and calculate the Intraclass Correlation Coefficient (ICC). The estimated ICCs indicate substantial taste stability: Engaging in sports exhibits an ICC of 0.80, reading and book interests 0.70, outdoor activities 0.61, movies 0.57, video games 0.54, and music 0.43 (the lower variance in music likely reflects ceiling effects in this population, where over 75 % of respondents consistently select “Very much”).

Table 3: Longitudinal Stability and Test-Retest Consistency of Cultural Tastes across Waves

Domain	Ego Self-Reported Tastes					Alter Perceived Tastes	
	ICC	r_{adj}	Exact (%)	± 1 (%)	$r_{W1 \rightarrow W3}$	r_{alter}	± 1 (%)
Sports	0.80	0.80	74.4	98.8	0.81	0.77	96.2
Books	0.70	0.72	76.9	99.1	0.65	0.70	95.8
Outdoor	0.61	0.62	68.8	98.8	0.62	0.63	96.0
Movies	0.57	0.57	76.3	99.4	0.62	0.54	97.8
Games	0.54	0.55	61.4	96.6	0.51	0.54	94.6
Music	0.43	0.50	78.5	99.7	0.36	0.64	98.1

^a ICC (Intraclass Correlation Coefficient) estimated from two-level random intercept models decomposing between-ego versus within-ego variance across Waves 1–3. r_{adj} , Exact, and ± 1 report mean adjacent wave-to-wave test-retest correlations and agreement percentages. $r_{W1 \rightarrow W3}$ reports multi-year stability across a 1.5-year span. Alter stability metrics are computed across consecutive waves for persisting dyads ($N = 2,546$).

Second, adjacent wave-to-wave test-retest correlations (r_{adj}) average between 0.50 and 0.80, with exact response agreement ranging from 61.4 % to 78.5 % and agreement within plus or minus one category exceeding 96.6 % to 99.7 % across all domains. Multi-year test-retest correlations spanning 1.5 years (Wave 1 to Wave 3; $r_{W1 \rightarrow W3}$) range from 0.51 to 0.81 for games, movies, outdoor activities, books, and sports. Third, for persisting ties, egos’ longitudinal perceptions of their alters’ tastes exhibit strong test-retest reliability ($r_{\text{alter}} = 0.54$ to 0.77), with 94.6 % to 98.1 % of perceptions remaining within plus or minus one scale step across consecutive waves.

3.2.4 Relational and Structural Controls

Subjective Closeness: Measured with the name interpreter, “How close are you to this person?” Response categories were “Not close” (10.1 %), “Somewhat close” (46.8 %), and “Very close” (43.0 %). This is treated as a factor variable in the models. As Marsden and Campbell (1984) noted in their classic paper, this item is the most reliable indicator of tie strength (Granovetter, 1973).

Relationship Type: Derived from the role relation name interpreter “How are you connected to this person?” I include binary indicators for whether the tie is a **Friend** (vs. other types) or a **Roommate/Dormmate** (capturing on-campus structural propinquity).

Frequency of Interaction: Measured with the name interpreter, “How often do you communicate with this person?” Responses were dichotomized into a **Daily** contact indicator (86.4 %) versus a reference category of **Weekly** contact (13.6 %).

Tie Duration: A continuous measure of how long the ego has known the alter (in years; Mean = 4.23 years, SD = 6.21), asked via the name interpreter: “How long have you known this person?” I include both the linear and squared terms (*Duration Squared*) to account for nonlinearities in the liability of newness.

Structural Embeddedness (Common Neighbors / Triadic Closure): To account for the collective, group-embedded nature of personal networks and control for triadic closure (Granovetter, 1973; Coleman, 1988), I construct a dyadic embeddedness metric from the perceived alter-by-alter acquaintance matrix. In each survey wave, respondents evaluated whether their nominated contacts knew one another (“As far as you know, does [Alter Name] know any of your other contacts that are listed below?”). From this dyadic matrix, I compute the number of common neighbors (shared contacts) that Alter j possesses with other alters in ego’s network (Mean = 5.74, SD = 4.18; ranging from 0 to 19). Because the ego is tied to every alter, each reported alter-alter connection directly represents a closed triangle (Ego – Alter $_j$ – Alter $_k$). In the models, this measure is standardized. For wave intervals where the alter-alter matrix was not included (Wave 6), the variable is standardized using a nonparametric wave-transition fixed effect to absorb the baseline shift, and sensitivity checks are conducted on the fully measured waves.

Demographics: I control for ego’s gender identification (46.0 % Women, 54.0 % Men) and perceived (from ego’s point of view) alter gender (49.0 % Women, 51.0 % Men), along with a multiplicative interaction term to capture same-gender vs. cross-gender homophily and heterophily (respectively). I also include a measure of *Race Homophily*, constructed by matching the ego’s self-reported race to the alter’s perceived race (27.0 % Homophilous, 9.1 % Heterophilous, 63.9 % Unknown alter race).

Time Period: Dummy variables for each wave transition interval (Wave 1 to 2, Wave 2 to 3, Wave 3 to 4, Wave 4 to 5, Wave 5 to 6, Wave 6 to 7, Wave 7 to 8) to allow the baseline hazard of tie decay to vary non-parametrically across the entire panel span.

3.3 Analytical Strategy

Because the data track relationships over time and individuals can nominate multiple alters, the data structure consists of dyad-periods nested within egos. To properly analyze tie decay, capture the high baseline volatility of personal networks (Cornwell et al., 2021; Weiss et al.,

2022), and account for right-censoring, I employ a discrete-time event history framework (Singer and Willett, 1993). Since the outcome variable—whether a tie is protected from decay into the next wave—is binary, I estimate protection from tie decay (the conditional probability of tie persistence given that the tie was active in the previous wave) using mixed-effects logistic regression.

This modeling strategy allows me to explicitly model the baseline hazard rate of tie decay across each wave transition, isolating the protective effects of cultural matching against this background volatility. Furthermore, because the discrete-time event history framework dynamically measures cultural matching at wave t to predict the hazard of tie decay into wave $t + 1$, the analytical model does not presuppose taste invariance within the observation period; rather, it evaluates whether contemporaneous cultural matching at the start of each transition interval protects the tie over the subsequent period.

To address the non-independence of observations arising from the clustered nature of the ego-network data (multiple ties per ego), I include random intercepts for egos (Agresti, 2007). This accounts for unobserved, time-invariant individual heterogeneity in the baseline propensity to maintain ties.

The models take the following general form:

$$\text{logit}(P(Y_{ijt} = 1 \mid \text{active at } t)) = \alpha_t + \beta_1 X_{ijt} + \beta_2 C_{ijt} + u_i \quad (1)$$

where Y_{ijt} is the binary indicator of protection from tie decay ($1 =$ persisted into wave $t + 1$; $0 =$ tie decay) for ego i and alter j at time t , α_t represents the period-specific baseline hazard intercept (captured via wave transition fixed effects), X_{ijt} represents the vector of core cultural matching predictors (closed-form cultural matching, open-ended activity matching, and cultural network opacity), C_{ijt} is a vector of time-varying dyadic and structural control variables (including subjective closeness, contact frequency, relationship type, race homophily, gender composition, tie duration, and structural embeddedness), and u_i is the ego-specific random intercept, assumed to be normally distributed with mean zero and variance estimated from the data ($u_i \sim \mathcal{N}(0, \sigma_u^2)$).

Because the sample was recruited from an undergraduate student cohort, a small proportion of nominated alters are also study participants (egos). In the full analytic panel of $N = 5,584$ dyad-periods (3,134 unique alters across 182 unique egos), exactly 102 unique alters (3.3 % of all nominated alters, representing 7.0 % of dyad-period observations) are also egos in the sample. Within the complete-case sample used for model estimation ($N = 5,349$ dyad-periods across 180 egos), this overlap corresponds to 374 dyad-periods (also 7.0 %). To ensure that this partial overlap does not violate the standard hierarchical nesting assumption, I also estimate *cross-classified multilevel models* (CCMM) with two-way crossed random effects:

$$\text{logit}(P(Y_{ijt} = 1 \mid \text{active at } t)) = \alpha_t + \beta_1 X_{ijt} + \beta_2 C_{ijt} + u_i + v_j \quad (2)$$

where $u_i \sim \mathcal{N}(0, \sigma_u^2)$ captures ego-specific heterogeneity and $v_j \sim \mathcal{N}(0, \sigma_v^2)$ captures alter-specific popularity or retention propensity, alongside models clustering on unique undirected dyads ($i \leftrightarrow j$).

All models are estimated via maximum likelihood using the `glmer` function from the `lme4` package (Bates et al., 2015) in R (R Core Team, 2026). In models that interact subjective closeness with cultural matching variables, I calculate and plot predicted probabilities to

facilitate the interpretation of the non-linear interaction terms (Mize, 2019). In addition, to ensure that findings are not sensitive to the multi-spell setup, two-way dyadic clustering, or between-ego selection, I evaluate three complementary sensitivity checks: an *absorbing first-decay* mixed-effects model restricting follow-up strictly to the initial continuous tie spell, *cross-classified and dyadic random effects models*, and an *ego fixed-effects* conditional logit model isolating within-ego variation.

4 Results

I present the findings in three stages. First, I examine the baseline main effects of cultural matching and opacity on protection from tie decay. Second, I test the competing moderation predictions (Hypotheses 3a and 3b) by interacting matching variables with subjective closeness. Finally, I evaluate the sensitivity of the findings by considering different specifications of the tie decay process that eliminate the possibility of tie rebirth, statistically evaluating the issue of egos who are also alters of other egos, accounting for survivorship bias due to differential sample retention of egos, and adjusting for all time-invariant ego-level characteristics via fixed effects (conditional logit) models.

4.1 Main Effects: Matching and Embeddedness

Table 4 presents the odds ratios from a series of mixed-effects logistic regression models predicting protection from tie decay. As standard network theory would predict, ties that are subjectively closer ($OR = 2.39, p < 0.001$ for “Close” ties in Model 1) are significantly more protected from tie decay. Interestingly, having daily contact ($OR = 1.12, p > 0.05$) does not reach conventional significance levels when adjusting for closeness and other controls in this sample.

Building on this structural baseline, Model 1 shows that *closed-form cultural matching* significantly protects ties from decay. For each additional broad cultural domain in which ego and alter cultural match, the odds of protection from tie decay increase by about 9.6 % ($OR = 1.10, p < 0.01$), net of controls.

In Model 2, I introduce *open-ended activity matching*. Cultural matching in specific leisure activities provides additional protective value against tie decay: each shared activity increases protection against tie decay by 8.0 % ($OR = 1.08, p < 0.05$). When both are included, closed-form cultural matching remains significant ($OR = 1.08, p < 0.05$), indicating that specific activity matching provides protective value beyond broad domain matching.

Model 3 tests the independent main effect of *cultural network opacity*—measured as the number of domains in which the ego reported “Don’t Know” regarding their alter’s preferences. I find that opacity predicts a 5.3 % increase in the hazard of tie decay per unknown domain ($OR = 0.95$ for protection from decay). However, this main effect is not statistically significant at conventional levels ($p > 0.05$) when adjusting for both closed-form and open-ended cultural matching simultaneously. The attenuation of this coefficient in the pooled mixed-effects model is likely due to the built-in, between-person correlation between reporting “Don’t Know” and having fewer known cultural matches. However, as I show

in the interaction models and the fixed-effects sensitivity check below, opacity remains a substantively significant liability for tie durability when considering only within-ego variance.

Finally, Model 4 introduces *structural embeddedness*—the standardized count of common neighbors (shared alter contacts) that an alter possesses within ego’s nominated network. Consistent with triadic closure and dynamic Simmelian tie arguments (Burt, 2000; Krackhardt and Handcock, 2007), having shared network contacts strongly and significantly protects ties from decay ($OR = 1.12, p < 0.05$): each standard deviation increase in the number of common alter contacts increases protection from tie decay by 12.2 %.

Importantly, adjusting for structural embeddedness does not eliminate the independent contribution of cultural matching. Both open-ended activity matching ($OR = 1.07, p < 0.05$) and closed-form cultural matching ($OR = 1.07, p = 0.057$) maintain their positive protective associations against tie decay. This shows that dyadic cultural matching is not an artifact of being embedded in dense peer cliques; rather, cultural matching and triadic structure operate as complementary, parallel anchors protecting ties from decay. Supplementary models examining the interaction between cultural matching and structural embeddedness revealed no significant moderation ($p = 0.65$), indicating that cultural matching protects ties from decay for both Simmelian and non-Simmelian ties.

To contextualize these coefficients, Figure 1 plots the average marginal effect of each core theoretical variable on the probability of protection from tie decay. Across the board, cultural matching serves as a protective mechanism against tie decay (positive estimates), whereas cultural network opacity accelerates tie decay (negative estimates).

4.2 Moderation by Subjective Closeness

While the main effects show that cultural matching generally protects ties from decaying, I next assess whether the protective effect is observed across all relationship types. To do so, I estimated auxiliary interaction models (not shown in the tables but visualized using predicted probabilities) to test whether the protective effects of cultural matching and the detrimental effects of cultural network opacity are moderated by baseline subjective closeness. Because interaction terms in logistic models can be opaque due to the non-linear link function, I visualize this critical interaction to make the core finding directly interpretable. Figure 2 plots the average marginal effects of the three cultural variables across ties rated as “Not Close,” “Somewhat Close,” and “Close.”

The results reveal a structured asymmetry in how cultural matching and cultural network opacity protect ties from decay conditioned on tie strength. I find a strong interaction for both closed-form cultural matching and cultural network opacity, but little interaction for open-ended activity matching. For broad, closed-form cultural matching and network opacity, the predictive slopes (marginal effects) are steepest and most significant for *strong* ties (those rated “Close”). For these stronger relationships, an increase in cultural matching substantially increases the predicted probability of protection from tie decay. Correspondingly, cultural network opacity imposes its most severe, statistically significant penalty precisely on these same strong ties: a lack of mutual cultural knowledge significantly accelerates their decay for ties rated as subjectively close. Conversely, for weak ties (those rated “Not Close”), the marginal effects for both closed-form cultural matching and opacity are relatively flat, indicating an intrinsic vulnerability to tie decay regardless of cultural matching or opacity in

Table 4: Odds Ratios for Protection from Tie Decay (Main Effects Models)

	Model 1 Closed	Model 2 + Open	Model 3 + Opacity	Model 4 + Embed.
Closed-Form Matches	1.096** (0.035)	1.083* (0.035)	1.070* (0.036)	1.067+ (0.036)
Open-Ended Matches		1.080* (0.034)	1.076* (0.034)	1.071* (0.033)
Network Opacity			0.947 (0.038)	0.952 (0.038)
Structural Embeddedness (Common Alters)				1.122* (0.050)
Subjective Closeness: Somewhat	1.440** (0.203)	1.388* (0.197)	1.368* (0.195)	1.342* (0.191)
Subjective Closeness: Close	2.388*** (0.359)	2.244*** (0.342)	2.188*** (0.336)	2.097*** (0.324)
Roommate/Dormmate	0.788 (0.195)	0.787 (0.194)	0.783 (0.194)	0.791 (0.197)
Friend	0.714 (0.175)	0.705 (0.173)	0.706 (0.174)	0.712 (0.177)
Frequency: Daily (vs Weekly)	1.119 (0.146)	1.107 (0.145)	1.085 (0.143)	1.081 (0.143)
Race Homophily: Same Race	0.950 (0.122)	0.944 (0.122)	0.935 (0.121)	0.932 (0.120)
Tie Duration (Scaled)	0.796 (0.183)	0.790 (0.182)	0.768 (0.178)	0.740 (0.172)
Tie Duration Sq (Scaled)	1.238 (0.303)	1.262 (0.309)	1.288 (0.317)	1.339 (0.331)
Num.Obs.	5349	5349	5349	5349
AIC	5147.9	5143.6	5143.7	5139.0
BIC	5273.0	5275.3	5282.0	5283.8

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: All models control for ego and alter gender, non-linear tie duration, and baseline hazard variations across all 7 wave transitions (coefficients omitted for space). Random intercepts for egos are included.

these broad domains.

This interaction suggests that cultural matching and mutual cultural knowledge constitute the substantive “core” of deeper friendships, requiring ongoing cultural matching to protect them from decay. Weak ties, on the other hand, are fleeting and superficial; their baseline hazard of decay is so high that cultural matching does little to prevent their decay. In contrast, the protective effect of specific, open-ended activity matching operates more uniformly across relationships of all strengths, providing a consistent anchor protecting ties from decay, whether the tie is new or well-established.

4.3 Sensitivity Analyses: Absorbing First Decay, Cross-Classified Models, and Ego Fixed Effects

To ensure that these empirical findings are not artifacts of model specification, multi-spell assumptions, or clustering structures, I evaluate three complementary sensitivity checks.

First, in Model 1 of Table 7, I evaluate sensitivity to the handling of recurrent spells and tie rekindling by estimating an *absorbing first-decay* mixed-effects logistic regression model ($N = 4,522$ complete dyad-periods across 3,770 dyads). Under this specification, each dyad is followed strictly until its first non-nomination event ($Y_{ijt} = 0$) or right-censoring; all subsequent recurrent spells following a temporary lapse are excluded from the risk set. As shown in the first column of Table 7, restricting the sample to the initial continuous spell yields results substantively equivalent to those of the primary models. Closed-form cultural matching remains a positive and statistically significant predictor of protection from tie decay ($OR = 1.086, p = 0.029$), structural embeddedness continues to strongly protect ties from decay ($OR = 1.120, p = 0.021$), and subjective closeness maintains its strong protective effect ($OR = 2.002, p < 0.001$ for “Close” ties). This indicates that the findings do not depend on the inclusion of recurrent or rekindled spells.

Second, to address potential non-independence arising from alters who are also study egos (or alters nominated across multiple egos), Table 5 reports estimates from cross-classified multilevel models with two-way (ego and alter) crossed random effects (Column 2) and undirected dyad clustering (Column 3, 3,708 unique dyad clusters within the complete-case sample). As shown in Columns 2 and 3 of Table 5, explicitly modeling crossed alter-level or dyad-level random effects leaves the point estimates and standard errors for the core predictors virtually identical to the standard hierarchical specification (Column 1): open-ended activity matching remains significant ($OR = 1.070\text{--}1.071, p < 0.05$), closed-form cultural matching remains positive ($OR = 1.065\text{--}1.066, p = 0.06\text{--}0.07$), and structural embeddedness continues to strongly protect ties from decay ($OR = 1.120\text{--}1.121, p < 0.05$). Furthermore, completely excluding the 374 dyad-periods where alter is also a survey ego (Column 4, $N = 4,975$) yields identical conclusions ($OR = 1.069, p = 0.043$ for open matching; $OR = 1.140, p = 0.006$ for structural embeddedness).

Third, to guard against potential panel attrition or survivorship bias (node persistence), I re-estimated the models across restricted subsamples based on survey retention. When follow-up is strictly right-censored at the point of ego survey non-response ($N = 4,855$ complete wave-to-wave transitions), open-ended activity matching remains a strong and significant protector against tie decay ($OR = 1.094, p = 0.007$), structural embeddedness

Table 5: Sensitivity Analysis: Cross-Classified Random Effects and Dyadic Clustering Models

	Model 1 Standard	Model 2 Crossed	Model 3 Dyad	Model 4 Excl. Ego-Alters
Closed-Form Cultural Matching	1.067+ (0.034)	1.065+ (0.034)	1.066+ (0.034)	1.062+ (0.036)
Open-Ended Activity Matching	1.071* (0.031)	1.071* (0.032)	1.070* (0.032)	1.069* (0.033)
Cultural Network Opacity	0.952 (0.040)	0.953 (0.041)	0.954 (0.040)	0.947 (0.042)
Structural Embeddedness (Common Alters)	1.122* (0.045)	1.121* (0.046)	1.120* (0.045)	1.140** (0.047)
Subjective Closeness: Somewhat	1.342* (0.143)	1.329* (0.145)	1.328* (0.144)	1.377* (0.150)
Subjective Closeness: Close	2.097*** (0.154)	2.071*** (0.157)	2.068*** (0.156)	2.161*** (0.163)
Roommate/Dormmate	0.791 (0.249)	0.798 (0.255)	0.791 (0.252)	0.732 (0.256)
Friend	0.712 (0.248)	0.708 (0.254)	0.713 (0.251)	0.740 (0.254)
Frequency: Daily (vs Weekly)	1.081 (0.132)	1.070 (0.134)	1.076 (0.133)	1.023 (0.138)
Tie Duration (Scaled)	0.740 (0.232)	0.745 (0.236)	0.742 (0.234)	0.666+ (0.242)
Tie Duration Sq (Scaled)	1.339 (0.247)	1.328 (0.252)	1.336 (0.250)	1.512 (0.256)
Ego Random Intercept SD (σ_u)	0.610	0.614	0.607	0.657
Alter/Dyad Random Intercept SD		0.427	0.328	
Dyad-Periods (N)	5349	5349	5349	4975
Unique Egos	180	180	180	180

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: All models control for ego and alter gender identity, gender interaction, race homophily, structural embeddedness, and wave transition fixed effects. Model 1 is the primary hierarchical random-intercept model. Model 2 estimates crossed random intercepts for both egos and alters. Model 3 specifies random intercepts for egos and unique undirected dyads. Model 4 excludes the 374 dyad-periods where alter is also a survey ego, within the complete-case modeled sample.

preserves tie durability ($OR = 1.087, p = 0.078$), and cultural network opacity accelerates tie decay ($OR = 0.928, p = 0.074$). Subsetting further to high-retention egos who completed 4 or more waves ($N = 4,605$ dyad-periods across 151 egos) and 6 or more waves ($N = 3,253$ dyad-periods across 92 egos) yields substantively identical results ($OR = 1.079, p = 0.028$ and $OR = 1.075, p = 0.076$ for open-ended matching, respectively).

Fourth, to evaluate whether the protective effect of cultural matching is driven by shared positive passions versus shared disinterest (negative homophily), Table 6 decomposes closed-form matching into positive interest alignment and shared disinterest. As shown in Model 2 of Table 6, exact positive interest matches significantly protect ties from decay ($OR = 1.072, p = 0.047$), and strong positive matches (Model 3, where both ego and alter rate interest as “Very interested”) provide an even stronger protective association ($OR = 1.087, p = 0.036$). In contrast, shared disinterest (e.g., both reporting “Not at all interested” in a domain) exhibits no statistically significant association with tie protection ($OR = 1.075, p = 0.386$ in Model 2; $OR = 1.069, p = 0.426$ in Model 3). This indicates that active positive cultural choices serve as a relational anchor protecting ties from decay, complemented by open-ended activity matching, which, by design, captures shared participation in favorite leisure pursuits ($OR = 1.070, p = 0.030$).

Fifth, in Model 2 of Table 7, I estimate an *ego fixed-effects* model using conditional logistic regression (Graubard and Korn, 2011) across $N = 1,640$ informative dyad-periods within egos exhibiting variation in tie decay. By stratifying the likelihood function by ego, this approach leverages strictly within-ego variance—comparing the alters a single ego retains with those it drops. This stringent test effectively purges all unobserved, time-invariant ego characteristics (such as baseline sociability, introversion, or personal response tendencies toward the “Don’t Know” option).

As shown in the second column of Table 7, the within-ego conditional logit estimates are substantively similar to the main mixed-effects models. Open-ended activity matching remains a statistically significant predictor of protection against tie decay ($OR = 1.056, p < 0.05$), while closed-form cultural matching remains positive ($OR = 1.046, p < 0.10$). Crucially, in this strict within-ego specification, *cultural network opacity* emerges as a highly significant accelerator of tie decay ($OR = 0.920, p < 0.01$), reducing the odds of protection from tie decay by 8.0 % per unknown domain. This indicates that while opacity may correlate with overall cultural matching rates between individuals, *within* an individual’s personal network, relational ignorance regarding a specific alter is an independent contributor to tie decay. Dyadic controls—such as being subjectively “Close” ($OR = 1.949, p < 0.001$) and “Somewhat Close” ($OR = 1.409, p < 0.01$)—also maintain their strong protective associations against tie decay.

To better assess the magnitude of these within-ego relationships, Figure 3 plots the within-ego marginal effects on the probability of protection against tie decay across observed values of closed-form cultural matching, open-ended activity matching, and cultural network opacity, conditional on ego fixed effects (relative to a baseline of zero matches or unknown domains). Across an individual’s personal network, increasing closed-form cultural matching from 0 to 6 domain matches progressively increases protection from tie decay by up to 4.7 %, while increasing open-ended activity matching from 0 to 5 shared activities elevates protection by up to 4.6 %. In contrast, cultural network opacity imposes an escalating penalty on tie durability within personal networks: increasing the number of unknown domains from 0 to 6

Table 6: Sensitivity Analysis: Decomposing Shared Positive Interests versus Shared Disinterest

	Model 1 Aggregated	Model 2 Exact Likes	Model 3 High Likes
Closed-Form Cultural Matching (Aggregated)	1.067+ (0.034)		
Shared Positive Interests (Exact)		1.072* (0.035)	
Shared Strong Interests (High)			1.087* (0.040)
Shared Disinterest / Dislikes		1.075 (0.084)	1.069 (0.083)
Open-Ended Activity Matching	1.071* (0.031)	1.070* (0.031)	1.068* (0.031)
Cultural Network Opacity	0.952 (0.040)	0.955 (0.040)	0.947 (0.039)
Structural Embeddedness (Common Alters)	1.122* (0.045)	1.122* (0.045)	1.122* (0.045)
Subjective Closeness: Somewhat	1.342* (0.143)	1.343* (0.143)	1.335* (0.143)
Subjective Closeness: Close	2.097*** (0.154)	2.099*** (0.155)	2.064*** (0.155)
Roommate/Dormmate	0.791 (0.249)	0.788 (0.249)	0.787 (0.249)
Friend	0.712 (0.248)	0.710 (0.248)	0.714 (0.248)
Frequency: Daily (vs Weekly)	1.081 (0.132)	1.078 (0.132)	1.081 (0.132)
Tie Duration (Scaled)	0.740 (0.232)	0.736 (0.232)	0.732 (0.232)
Tie Duration Sq (Scaled)	1.339 (0.247)	1.344 (0.247)	1.352 (0.247)
Ego Random Intercept SD (σ_u)	0.610	0.608	0.609
Dyad-Periods (N)	5349	5349	5349
Unique Egos	180	180	180

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: All models include controls for ego and alter gender identity, gender interaction, race homophily, wave transition fixed effects, and ego random intercepts ($N = 5,349$ dyad-periods across 180 egos). Model 1 is the primary aggregated matching model. Model 2 decomposes closed-form matching into exact positive interest matches and shared disinterest. Model 3 isolates strong positive matches (both rating interest as “Very much”).

reduces protection from tie decay by 8.0 %. This indicates that even after conditioning out time-invariant ego characteristics and unobserved individual traits, cultural matching and opacity continue to predict the decay of social ties.

5 Discussion

5.1 Summary of Key Results

Using longitudinal ego-network data from the *NetSense* study, I evaluated the empirical implications of a cultural matching model of tie decay in personal networks. The results provide consistent evidence that cultural matching is an important mechanism protecting social ties from decay (Burt, 2000). Specifically, both broad, closed-form cultural matching and specific, open-ended activity matching independently reduce the hazard of tie decay, even after controlling for dyadic structural embeddedness (triadic closure with other alters). While structural embeddedness itself strongly protects against tie decay ($OR = 1.12, p < 0.05$), it does not displace the effect of cultural matching; rather, it serves as a complementary protective mechanism.

Furthermore, I identified cultural network opacity—the lack of mutual knowledge of others’ tastes and practices—as an independent cultural mechanism that accelerates tie decay. Crucially, I found that protection from tie decay via closed-form cultural matching and the accelerating effect of cultural network opacity are most pronounced among *strong* ties as measured by subjective closeness (Marsden and Campbell, 1984). For these affectively significant relationships, cultural matching serves as an important mechanism protecting ties from decay. Weaker, less affectively significant ties, on the other hand, have such an elevated baseline hazard of decay that cultural matching can do little to protect them from decaying. Notably, the protective effect of specific, *open-ended* activity matching—getting at more granular and specific types of cultural commonalities—against tie decay remains consistent across both weak and strong ties.

5.2 Limitations and Suggestions for Future Work

Despite these promising findings, several limitations point toward fruitful avenues for future research. First, regarding operationalization and measurement constraints, the current measurement approach for closed-form cultural matching aggregates across broad domains of lifestyle and taste, with little variance in their overall legitimacy and consecration. Future research could further disaggregate these domains to examine whether traditionally high-status cultural genres (e.g., classical music, highbrow literature) confer distinct relational advantages compared to more popular or subcultural domains (DiMaggio, 1987; Lizardo, 2006; Meuleman and Jæger, 2023). In addition, while the measures focus primarily on positive cultural matching and lack of knowledge (opacity), future work should systematically investigate the role of shared active cultural *dislikes* and mutual distastes as distinct symbolic boundaries in tie decay (Edelmann and Vaisey, 2014).

Second, the network measures capture ego’s *perceived* relational landscape. While ego-centric network data are uniquely suited for measuring an actor’s cognitive orientation,

Table 7: Sensitivity Models: Absorbing First Decay and Ego Fixed-Effects

	Absorbing First Decay	Ego FE (clogit)
Closed-Form Matches	1.086* (0.041)	1.046+ (0.027)
Open-Ended Matches	1.038 (0.036)	1.056* (0.025)
Network Opacity	0.963 (0.042)	0.920** (0.029)
Structural Embeddedness (Common Alters)	1.120* (0.055)	1.055 (0.036)
Subjective Closeness: Somewhat	1.254 (0.191)	1.409** (0.155)
Subjective Closeness: Close	2.002*** (0.336)	1.949*** (0.233)
Roommate/Dormmate	0.753 (0.201)	0.810 (0.149)
Friend	0.751 (0.196)	0.806 (0.149)
Frequency: Daily (vs Weekly)	1.154 (0.185)	1.078 (0.096)
Race Homophily: Same Race	1.030 (0.151)	0.926 (0.081)
Tie Duration (Scaled)	0.759 (0.187)	0.820 (0.146)
Tie Duration Sq (Scaled)	1.304 (0.346)	1.202 (0.220)
Num.Obs.	4522	1640
AIC	4209.0	10669.8
BIC	4350.2	10777.8

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: Model 1 restricts follow-up strictly to the initial continuous tie spell until first tie decay or censoring, excluding all subsequent recurrent/rekindled spells. Model 2 stratifies the likelihood by ego (conditional logit), isolating within-ego variation. Both models adjust for alter gender and wave transition fixed effects; Model 1 also includes ego gender and an ego random intercept.

subjective closeness, and perceived cultural knowledge about their alters, they capture only one side of the dyad. In non-reciprocated or unidirectional ties, relational orientations may be asymmetric, reflecting underlying differences in social status, popularity, or aspirational connection (Carley and Krackhardt, 1996). In such cases, an ego may actively work to maintain a culturally matched tie with a higher-status peer even if the alter does not reciprocate the nomination (McFarland et al., 2014). Although ego fixed-effects and dyadic controls account for an ego’s general nomination patterns, future investigations with complete dyadic reciprocity data can formally evaluate how cultural matching interacts with status differentials and reciprocal confirmation.

Third, an important question concerns the *temporal horizon* of cultural matching beyond the college context. The panel follows students through their undergraduate trajectory across semester intervals, capturing the critical period when campus ties are actively formed and pruned. However, a compelling open question is whether cultural matching continues to sustain social ties across major post-college life transitions—such as graduation, geographic dispersal, labor market entry, and family formation. When individuals no longer share the daily scaffolding of an institutional campus environment, does deep cultural matching provide the necessary conversational and ritual currency to protect distant ties from decay across decades? Alternatively, do enduring college ties persist primarily via accumulated relational history and institutional memory, rendering cultural matching less decisive once physical co-presence ends? Future long-term multi-decade panel studies will be vital for determining the life-course boundaries of cultural matching.

Finally, the empirical focus on a college-age cohort during emerging adulthood delineates specific institutional scope conditions for these claims. The transition to college represents a unique developmental and institutional ecology characterized by high baseline network churn, rapid relational renegotiation, and geographic concentration (McFarland et al., 2014). While this setting provides an ideal testing ground for observing tie decay and protection from decay in short- to medium-term time scales within a single ecology, future research should examine whether cultural matching operates similarly to protect ties from decay across different life stages and institutional domains. In particular, comparative studies in workplace organizations (McFarland et al., 2014; Kim et al., 2024), civic associations, or among older adults (Cornwell et al., 2008) could shed light on how the capacity of cultural matching to protect against tie decay varies across differing levels of structural constraint.

5.3 Implications: Cultural Capital and Relational Maintenance

These findings carry significant theoretical implications for understanding the relationship between tastes and networks, particularly regarding the role of tie strength. The results heavily favor the empirical implications of the strong-tie matching prediction (*Hypothesis 3a*) over the weak-tie scaffolding prediction (*Hypothesis 3b*).

Aligning with previous research (Edelmann and Vaisey, 2014; Vaisey and Lizardo, 2010), the longitudinal event history framework reveals that emerging friendships require ongoing cultural matching to sustain intimacy and prevent tie decay. It is the *strong* ties that are highly sensitive to cultural matching and opacity. Weak ties, suffering from a high liability of newness, rapidly decay with or without cultural matching. These results extend recent observations in life course and network research showing that while significant network

turnover is a ubiquitous feature of critical life transitions, individuals actively curate their core networks to maintain stable, supportive ties (Roy et al., 2022; Weiss et al., 2022; Kempnich et al., 2024; Vacchiano et al., 2024).

Returning to the puzzle posed in the introduction—how to account for durable cultural profiles in the face of volatile networks (Bourdieu, 1984; Holt, 1998)—these results help to close the loop. If persons use relatively stable cultural dispositions to selectively maintain their networks, it suggests that cultural capital is not merely a passive status marker or a “thin” effluvium of network position. Rather, it is an active navigational tool for managing social environments. People churn through weak ties at a high baseline rate, but they rely on their durable cultural compass to actively curate and protect the core relationships that matter from decay.

Furthermore, these findings speak directly to the classic distinction between choice homophily and induced homophily (McPherson et al., 2001). If cultural matching actively prevents tie decay and opacity accelerates it, then a substantial portion of the observed homophily in established social networks is likely due to selective protection against tie decay (choice homophily) rather than pure structural constraint or network-induced adaptation (Lizardo, 2006; Lewis and Kaufman, 2018). By protecting culturally matched ties from decay and shedding incompatible ones over time, networks become homophilous via an active evolutionary sorting process (Rivera et al., 2010).

While early network theories emphasized structural constraints or multiplexity (Hammer, 1985; Davis, 1963), the findings suggest that cultural matching serves as an additional micro-interactional mechanism that, alongside structural embeddedness, protects close relationships from decay. In the absence of mutual cultural knowledge (high opacity), even subjectively close ties face a higher hazard of decay. This developmental view of networks helps reconcile the volatility of personal networks with the relative stability of individual cultural profiles.

Beyond network theory, these findings carry practical implications for institutions aiming to foster social cohesion. For university administrators and community organizers seeking to reduce social isolation, these results underscore that physical proximity and shared structural affiliations are most effective when paired with opportunities for common cultural interests and activities to sustain durable relationships. Facilitating environments where individuals can discover and activate cultural matching—from niche clubs to diverse recreational programs—may be essential for transforming fleeting initial encounters into resilient, long-term support networks.

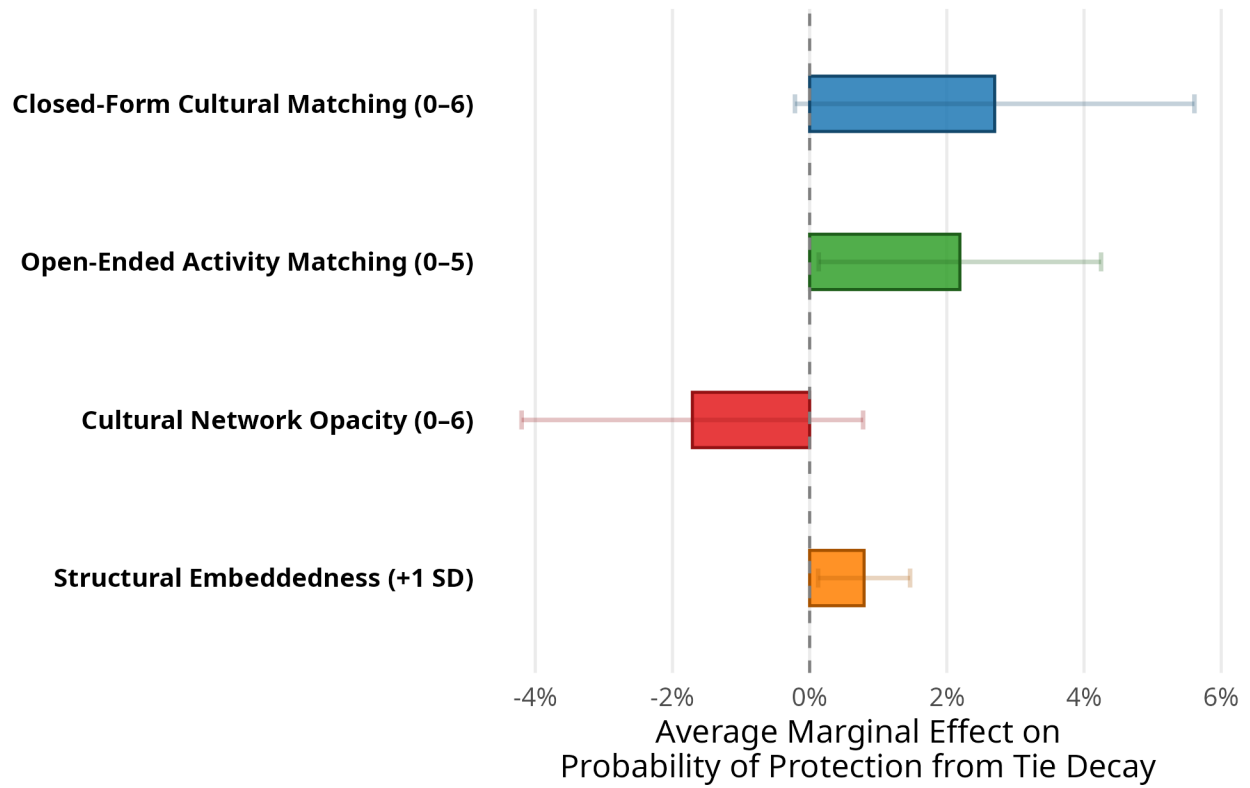


Figure 1: Predicted Probability of Protection from Tie Decay by Cultural Matching and Cultural Network Opacity.

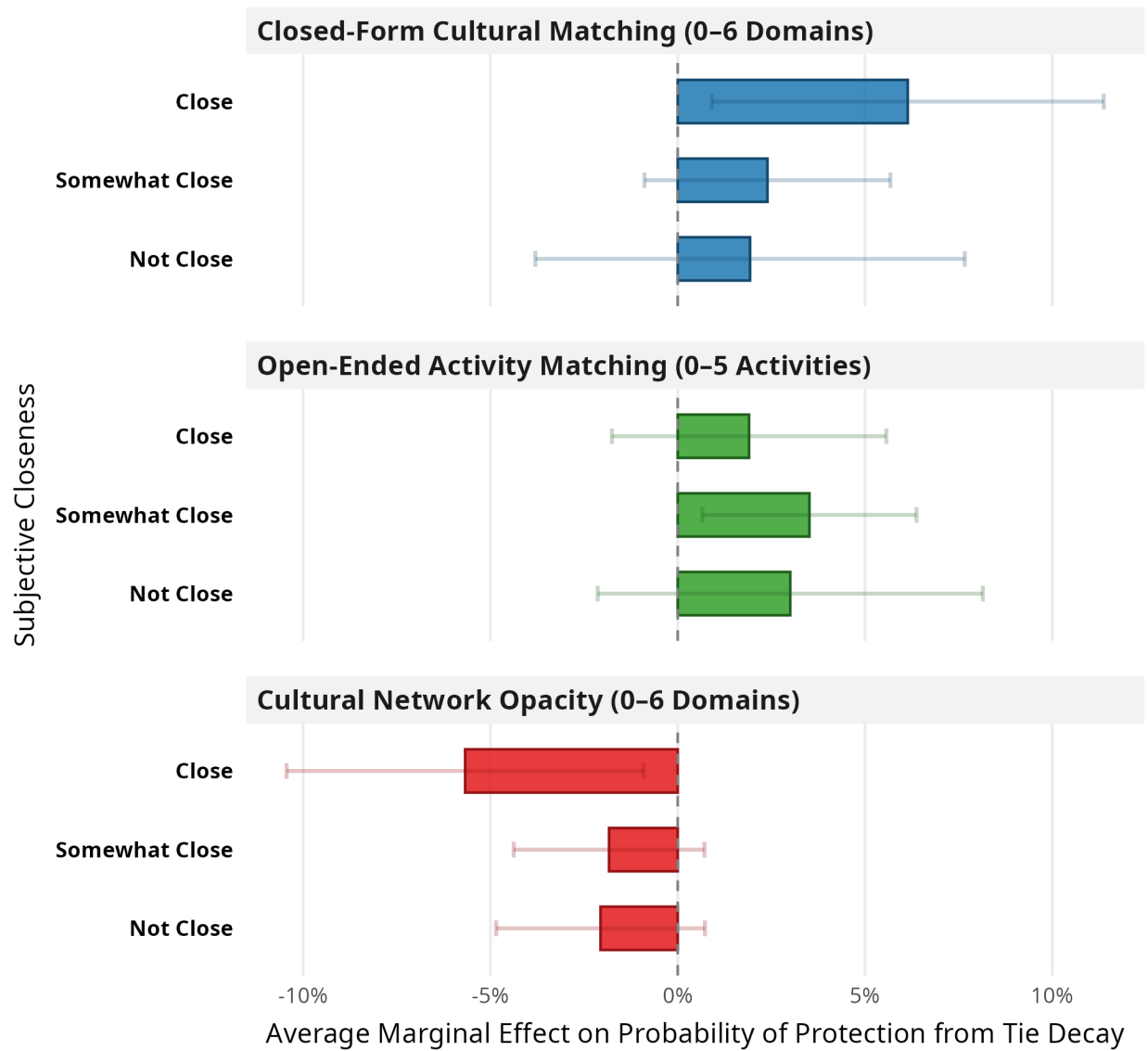


Figure 2: Predicted Probability of Protection from Tie Decay by Cultural Matching, Network Opacity, and Subjective Closeness.

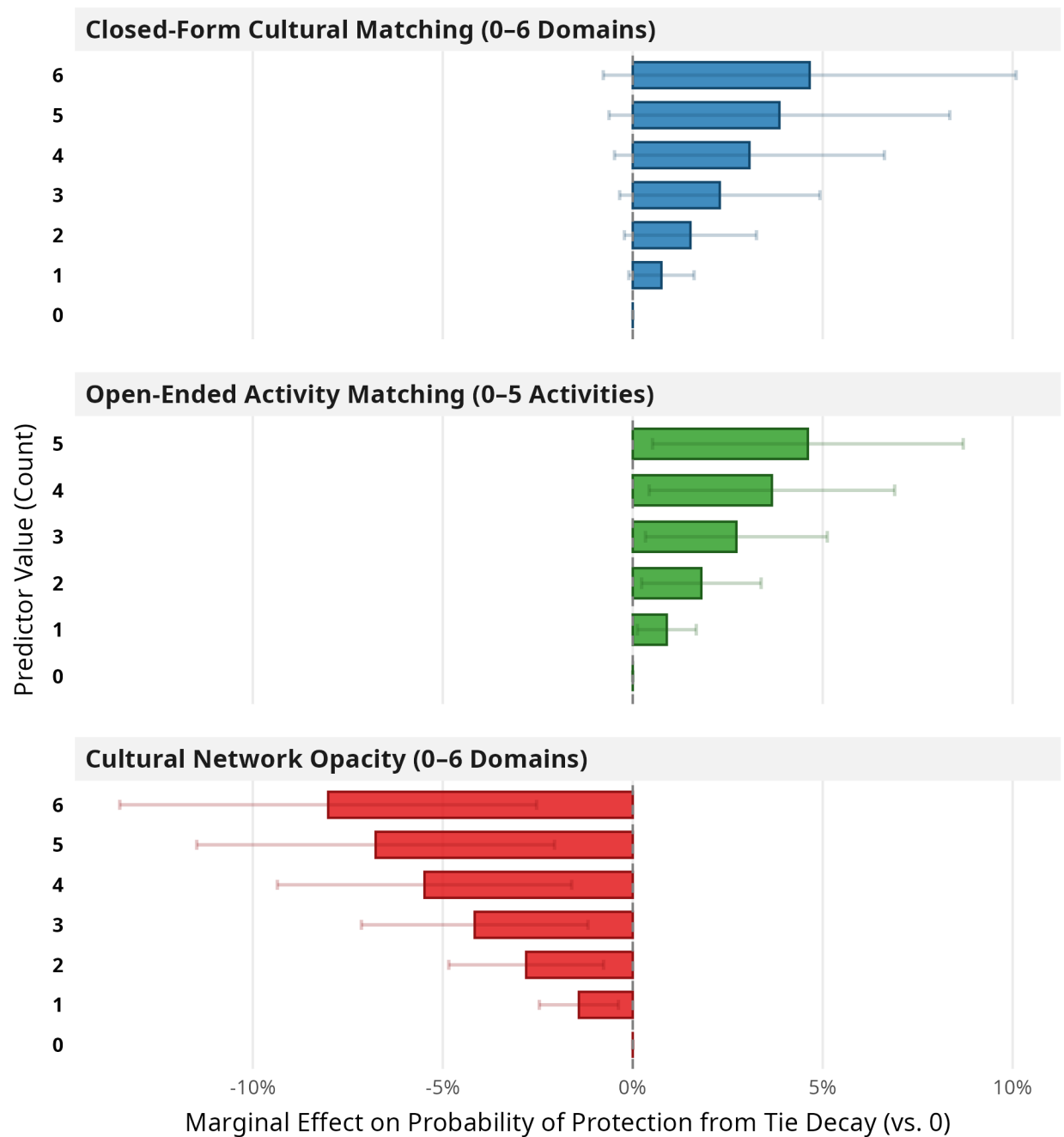


Figure 3: **Within-Ego Marginal Effects on Protection from Tie Decay by Cultural Matching and Opacity Counts (Ego Fixed-Effects).**

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